

Departamento de Eletrónica, Telecomunicações e  
Informática

# **Foundations of Machine Learning**

**LECTURE 1 : INTRODUCTION**

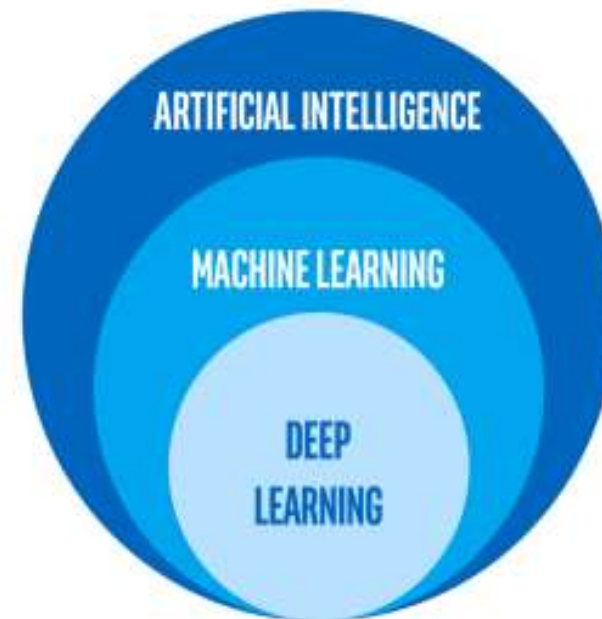
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# Artificial Intelligence (AI)

**AI** is a general purpose technology that may influence every industry (similar to electricity, internet) .

AI is based on

Machine Learning (ML) & Deep Learning (DL) algorithms



# PROGRAM

## **Supervised learning**

- Linear (univariate/ multivariate) regression
- Logistic regression. Regularization
- Artificial Neural Networks (ANN)
- Support Vector Machines (SVM)
- Decision Tree (DT);
- Naive Bayes classifier
- k-Nearest Neighbor (k-NN) classifier

## **Unsupervised learning**

- K-means clustering
- Data dimensionality reduction
- Principal components analysis (PCA)

## **Deep Learning**

Deep Learning architectures :

- CNN (Convolutional Neural Networks);
- LSTM (Long Short Term Memory) neural network
- Multivariate Gaussian approach for Anomaly Detection
- Recommender Systems

# Evaluation

Lectures & labs: 3 hours per week.

## **Practical component - 50% of the final grade**

Practical component consists of 2 projects, developed in a group of two students.

The first project is evaluated based on a submitted report (IEEE format) and a short (10-15 min.) oral presentation.

The second project is evaluated based on a submitted report (IEEE format).

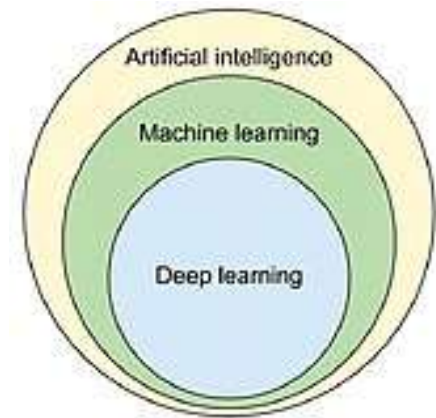
The students are encouraged to use Latex text editor.

Overleaf is a convenient platform for collaborative writing and publishing using Latex (<https://www.overleaf.com/>) .

## **Theoretical Component – 50% of the final grade (Final exam).**

# Why Machine/Deep Learning

- **Sensors** get cheaper (e.g. widely available IoT devices)
- Exponential **growth of data** – WSN/IoT, medical records, biology, engineering, etc.
- **Data sources**: sound, vibration, image, electrical signals, accelerometer, temperature, pressure, LIDAR etc.
- Increasing **computational resources**.
- **Complex Applications:**
  - ✓ Autonomous driving;
  - ✓ Intelligent robotics;
  - ✓ Computer Vision;
  - ✓ Natural Language Processing (Speech recognition, Machine translation)
  - ✓ 5G+ networks



# A bit of history

- **1950**, Alan Turing: "Computing Machinery and Intelligence" define the question "Can machines think?"  
=>Turing test.
- **1956** –The field of Artificial Intelligence (AI) formally established at the conference in Dartmouth College.
- **1959**, Arthur Samuel: “ Field of study that gives computers the ability to learn without being explicitly programmed ”.
- **1998**, Tom M. Mitchell: “ Can the computer program learn from experience ? ”.

# Machine Learning – “definition”

„A computer program is said to learn from experience  $E$  with respect to some task  $T$  and some performance measure  $P$ , if its performance on  $T$ , as measured by  $P$ , improves with experience  $E$ .“  
**(T. Mitchell 1998)**

- **Given**

- a task  $T$  (e.g. classify spam/regular emails)
- a performance measure  $P$  (weighted sum of mistakes)
- some experience  $E$  with the task (e.g. hand-sorted emails)

- **Goal**

- generalize the experience in a way that allows to improve the machine performance on the task

# Learning to classify documents



## Web page:

Company, Personal, University, etc.

## Articles:

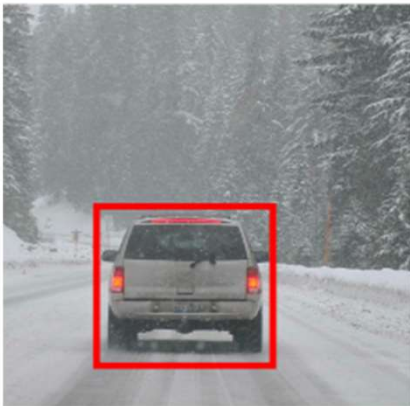
Sport, Political, History, etc.

# Computer Vision

Learning to detect & recognize faces



# Computer Vision Tasks

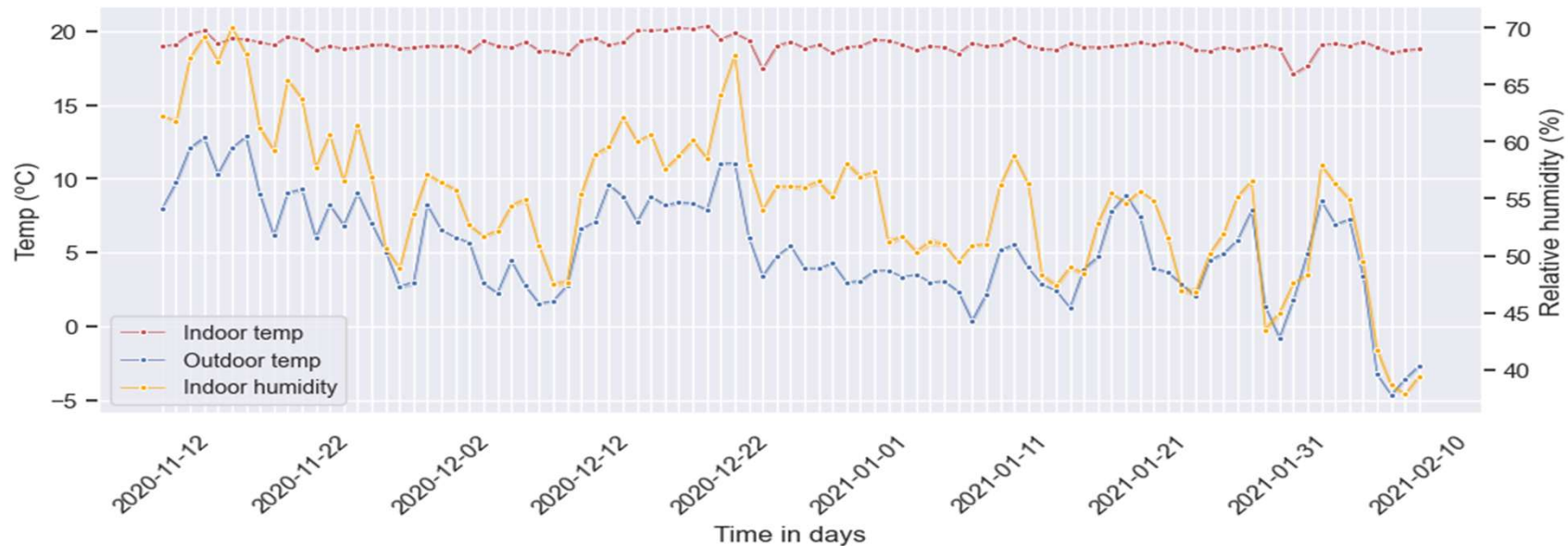
| Image classification                                                              | Classification & Localization                                                                              | Detection                                                                           |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  | <br>$b_x, b_y, b_h, b_w$ |  |

**Image classification:** input a picture into ML/DL model and get the class label (e.g. person, bike, car, background, etc.)

**Classification & localization:** the model outputs not only the class label of the object but also draws a bounding box (the coordinates) of its position in the image.

**Object Detection:** outputs the position and labels of several objects.

# Time Series (TS) Data

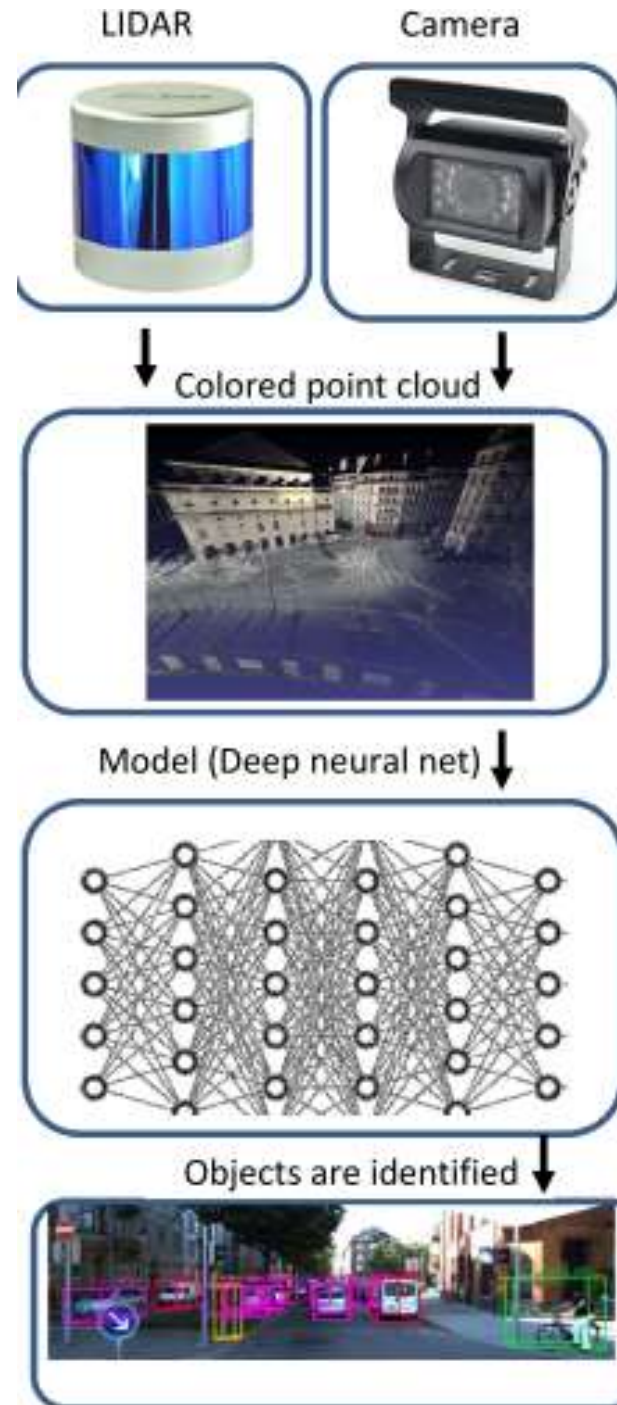


Time Series (TS) - collection of samples recorded at a sequence of time intervals

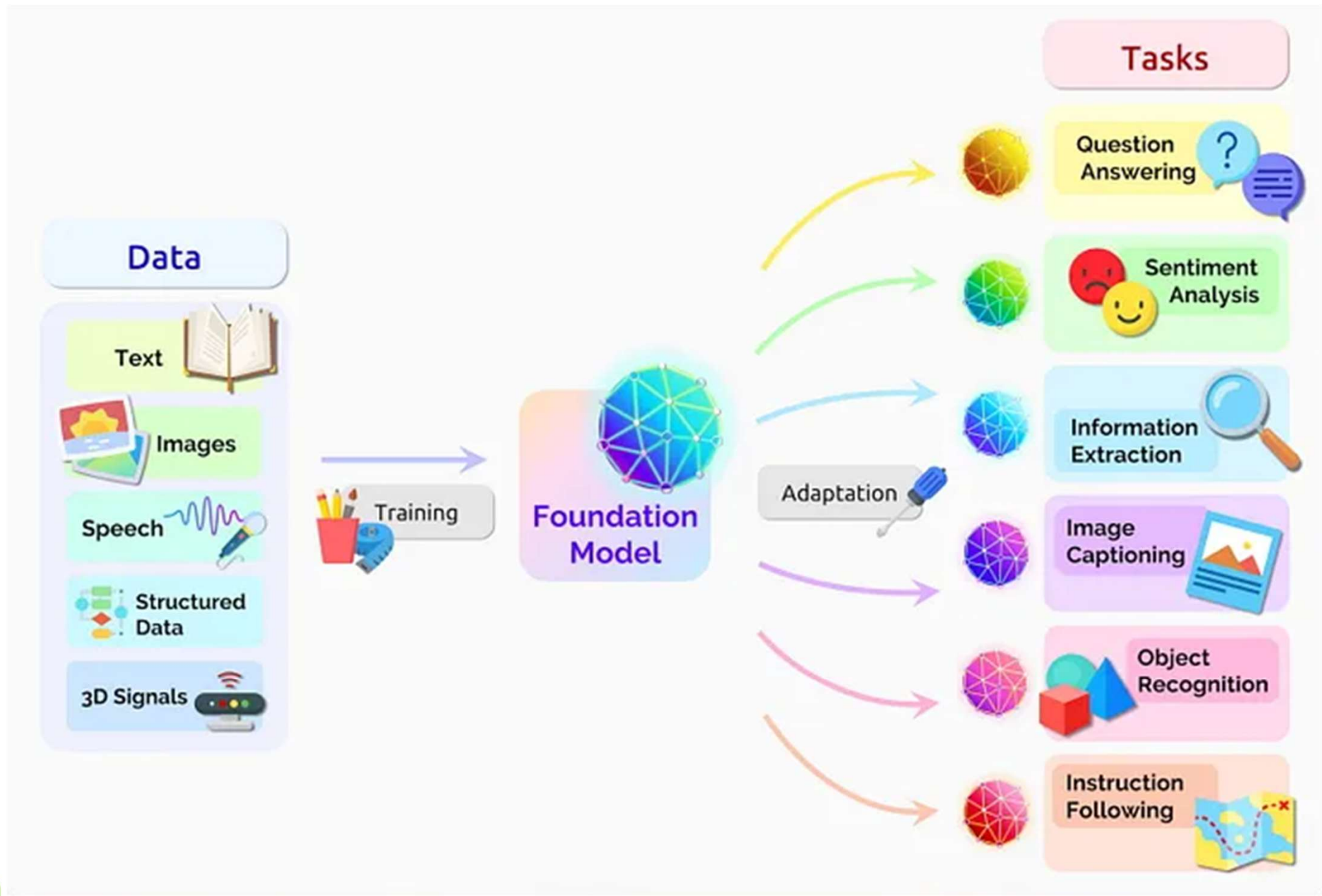
TS forecasting (prediction) => based on past samples, predict future trends, seasonality, anomalies, etc. Many applications:

- Key Performance Indicators (KPIs) : network traffic prediction
- Smart Homes – predict indoor temp., heating set-point, thermal comfort
- Weather forecast – heat waves, flooding
- WSN physical layer – channel modelling / estimation

# Multimodal Object Detection

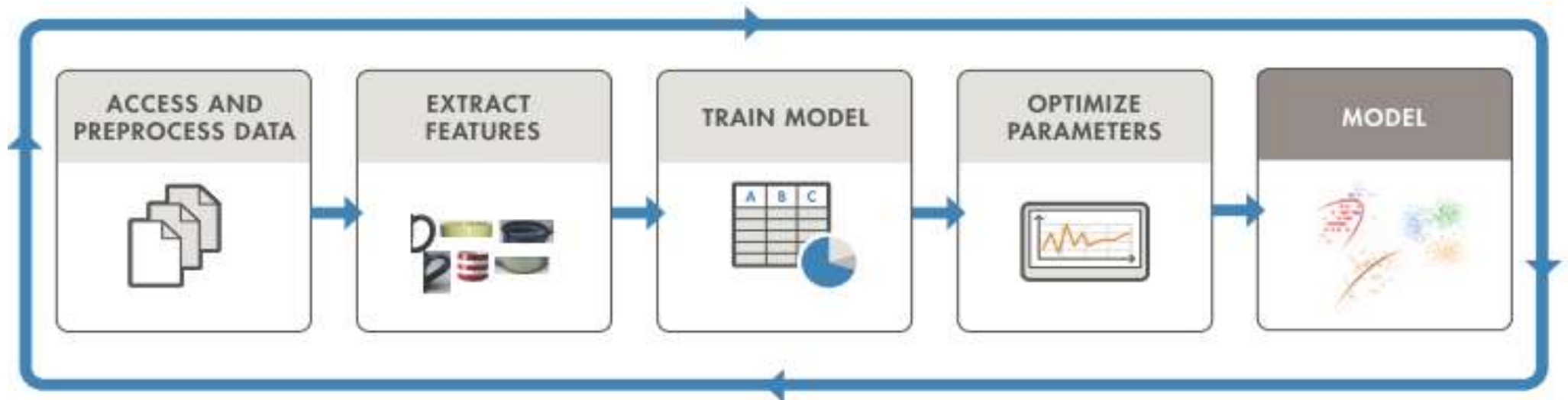


# Multimodal generative AI models

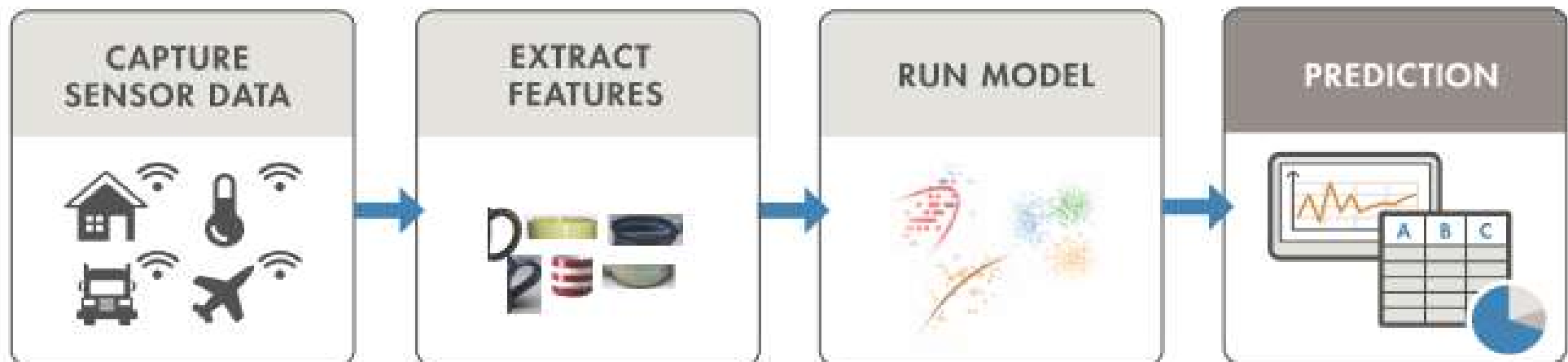


# ML workflow

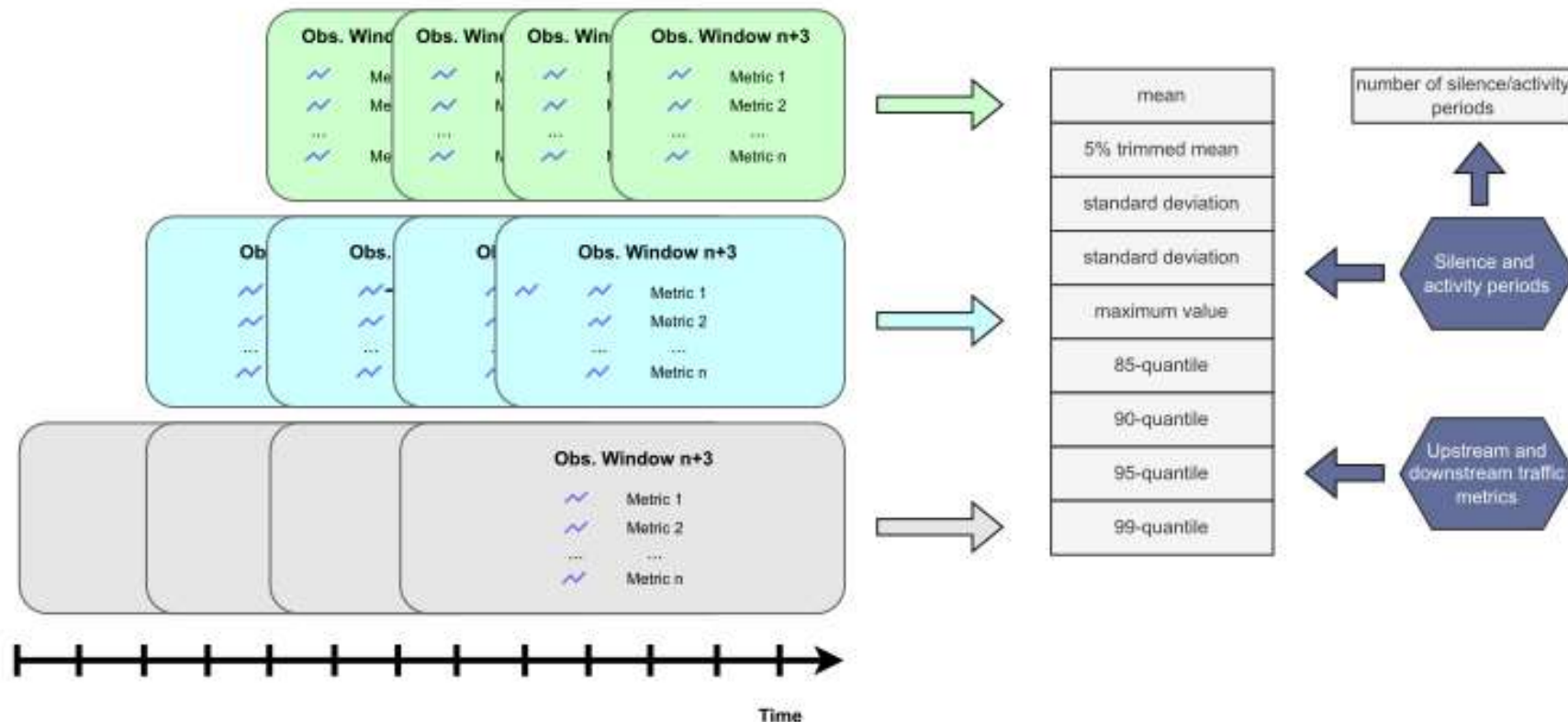
**Train:** Iterate until achieve satisfactory performance (**off-line**)



**Predict:** Integrate trained models into applications (**real time**)



# From Raw data to Hand-crafted features



## Raw data:

collected upstream/downstream network traffic metrics; sensor measurements  
uploaded packets (#, Bytes), downloaded packets (#, Bytes), silence/activity periods

## Feature extraction (input vector $\mathbf{x}$ ) - e.g. statistical metrics

mean, max, min, standard deviation, different quantiles, over multiple sub-windows

**Class (label  $\mathbf{y}$ )** : Network traffic OK (0) / NOT OK (1)

# Machine Learning Approaches

## **Supervised Learning**

Given examples with “correct answer” (labeled examples)  
(e.g. given dataset with spam/not-spam labeled emails)

## **Unsupervised Learning**

Given examples without answers (no labels).

## **Deep Learning**

Automatically extract hidden features (in contrast to hand-crafted features). Need a lot of data (Big data) . Need for very high computational resources (GPUs).

## **Reinforcement Learning**

On-line (on the fly) learning, by trial and error

Applications: intelligent robotics, autonomous systems

# Supervised Learning

Requires labeled data (examples with “correct answer”).

**Regression:** The Labels are real numbers.

**Ex.** Predict the house price (output) based on data for the house area and number of bedrooms (features).

| Living area (feet <sup>2</sup> ) | #bedrooms | Price (1000\$s) |
|----------------------------------|-----------|-----------------|
| 2104                             | 3         | 400             |
| 1600                             | 3         | 330             |
| 2400                             | 3         | 369             |
| 1416                             | 2         | 232             |
| 3000                             | 4         | 540             |
| ⋮                                | ⋮         | ⋮               |

**Classification:** The Labels are categorical values (class 1, class 2, etc.)

**Ex.** Predict normal (0) or abnormal (1) state of data center computers:

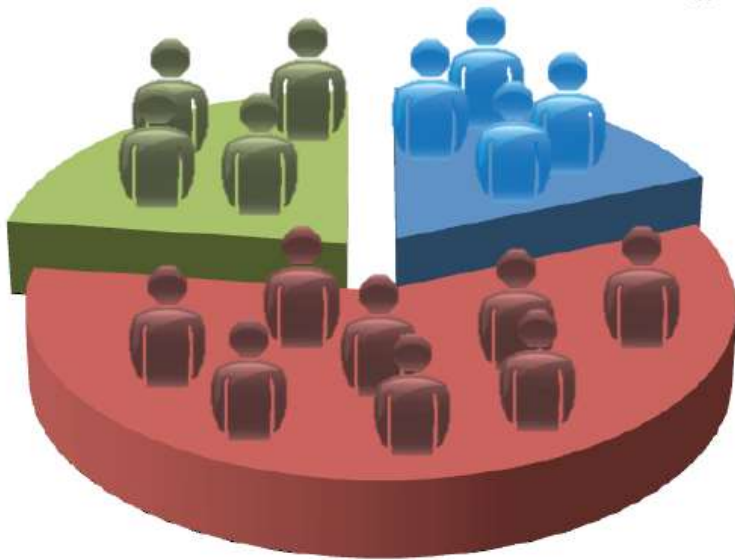
**Features:** memory use of computer ; number of disc accesses /sec; CPU load ; network traffic; silence

# Unsupervised Learning

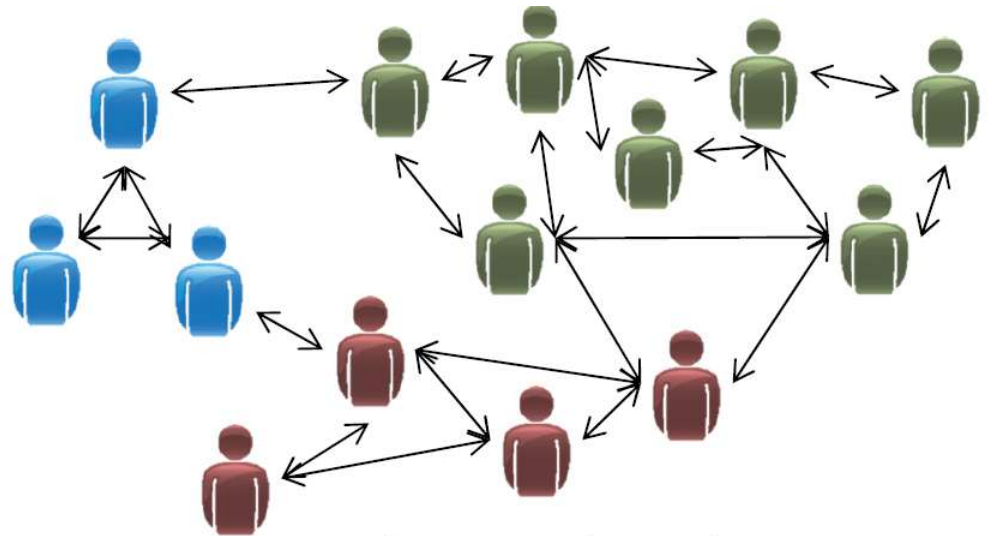
Given unlabeled data (NO answers)

**Features:** education, job, age, marital status, etc.

Market segmentation



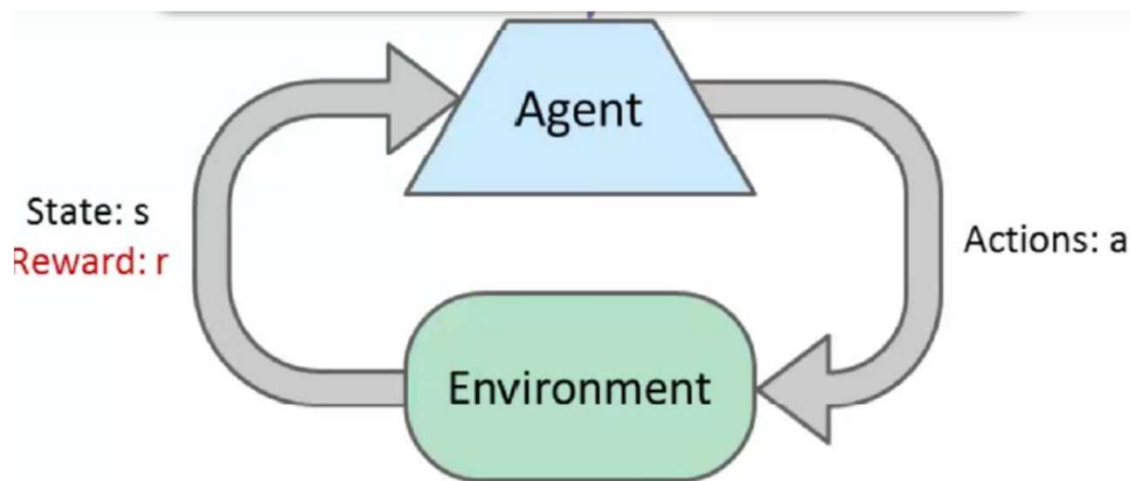
Social network analysis



**Clustering:** Given a collection of examples (e.g. user profiles with a number of features). Each example is a point in the multidimensional space of features. Find a similarity measure that separates the points into clusters.

**-K-means clustering**

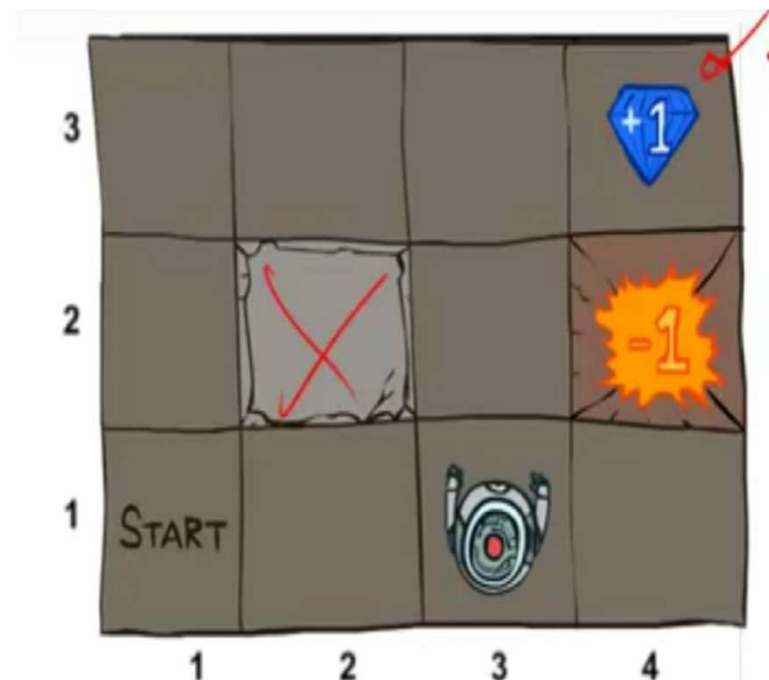
# Reinforcement Learning



On-line learning by taking actions  
and getting rewards/penalties.  
intelligent robotics =>

Learn to act so as to maximize  
expected rewards

Learning is based on observed episodes



# Why Deep Learning ?

Hardware get smaller.

Sensors get cheaper, widely available IoT devices with high sample-rate.

Data sources: sound, vibration, image, electrical signals, accelerometer, temperature, pressure, LIDAR, etc.

**Big Data:** Exponential growth of data, (IoT, medical records, biology, engineering, etc.)

How to deals with **unstructured data** (image, voice, text, EEG, ECG, etc.) =>  
What are the best feature ?

Deep Neural Networks: first extract (automatically) the hidden features, then solve ML tasks (classification, regression)

# DL for 5G+ networks

Data traffic forecast – a key mechanism to automate 5G Network

## What 5G is about



# Data Types

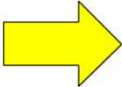
## 1. Numeric (Quantitative) features

- Integer numbers
- Floats (decimals) - temperature, height, weight, humidity, etc.

## 2. Boolean – True/False

## 3. Categorical features - gender, days of the week, seasons, country of birth, colors, etc.

How to deal with categorical features ? - One-hot encoding  
(1,0) transforms  $n$  categories into  $n$  features

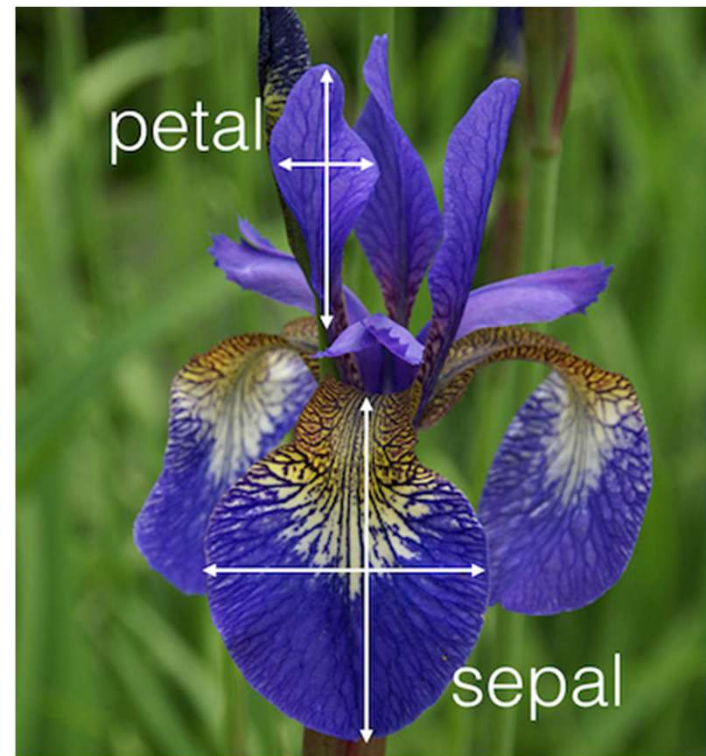


| Color  |
|--------|
| Red    |
| Red    |
| Yellow |
| Green  |
| Yellow |

| Red | Yellow | Green |
|-----|--------|-------|
| 1   | 0      | 0     |
| 1   | 0      | 0     |
| 0   | 1      | 0     |
| 0   | 0      | 1     |

# Iris Plant data

- Iris Plant data – benchmark dataset for illustration of ML methods.
  - UCI Machine Learning Repository  
<http://www.ics.uci.edu/~mlearn/MLRepository.html>
  - 3 flower types (classes):
    - Setosa
    - Virginica
    - Versicolour
  - 4 attributes (features)
    - Sepal width and length
    - Petal width and length

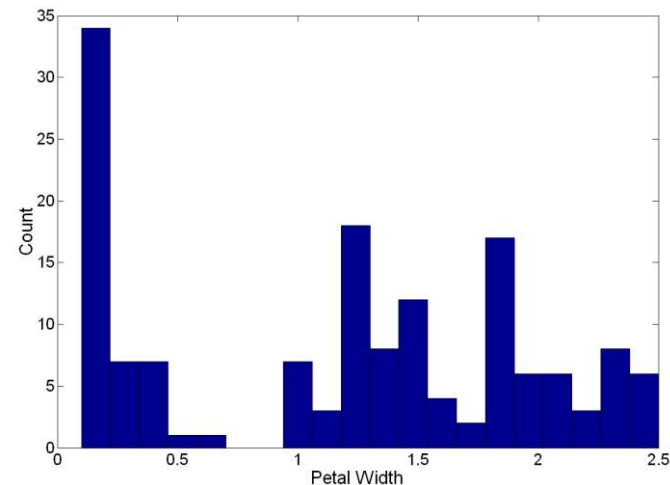
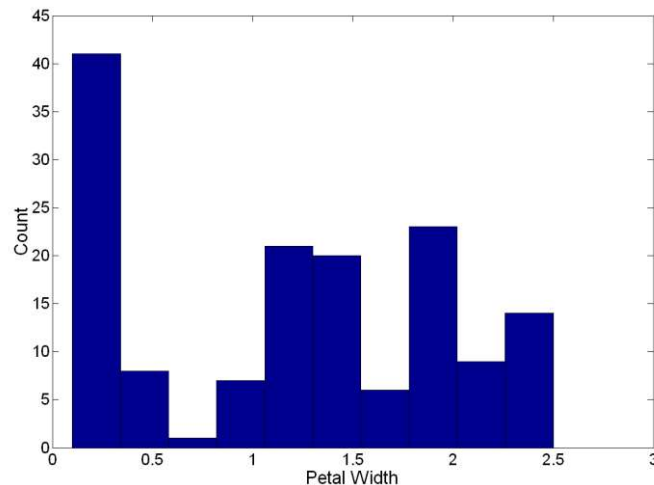


# Data Visualization (1)

- **Histograms**

- Show the distribution of values of a single feature
- Divide the range of values of a single feature into bins and show bar plots of the number of examples in each bin.
- Histogram shape depends on the number of bins

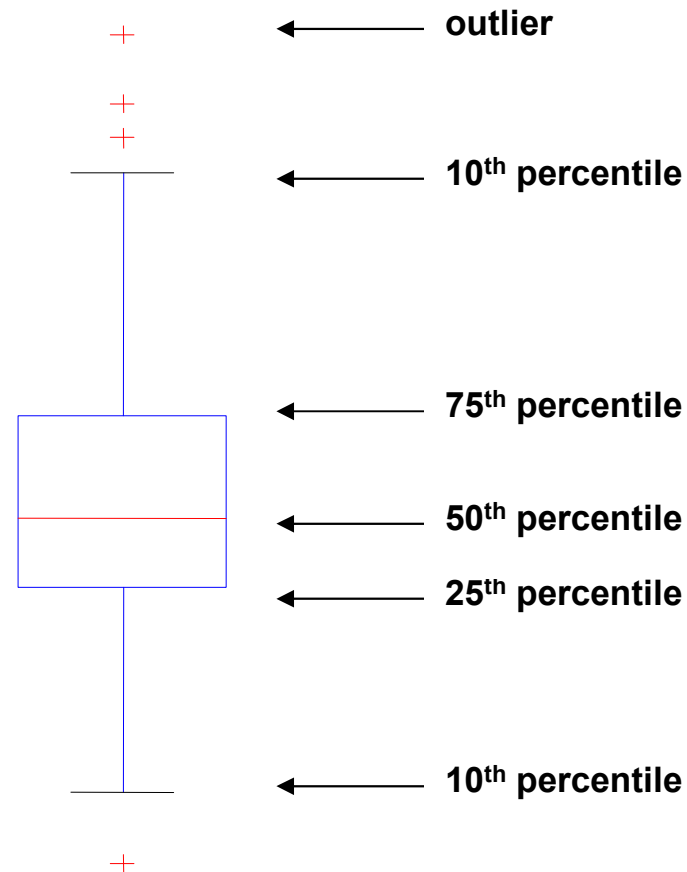
- Example: Petal Width (10 and 20 bins, respectively)



# Data Visualization (2)

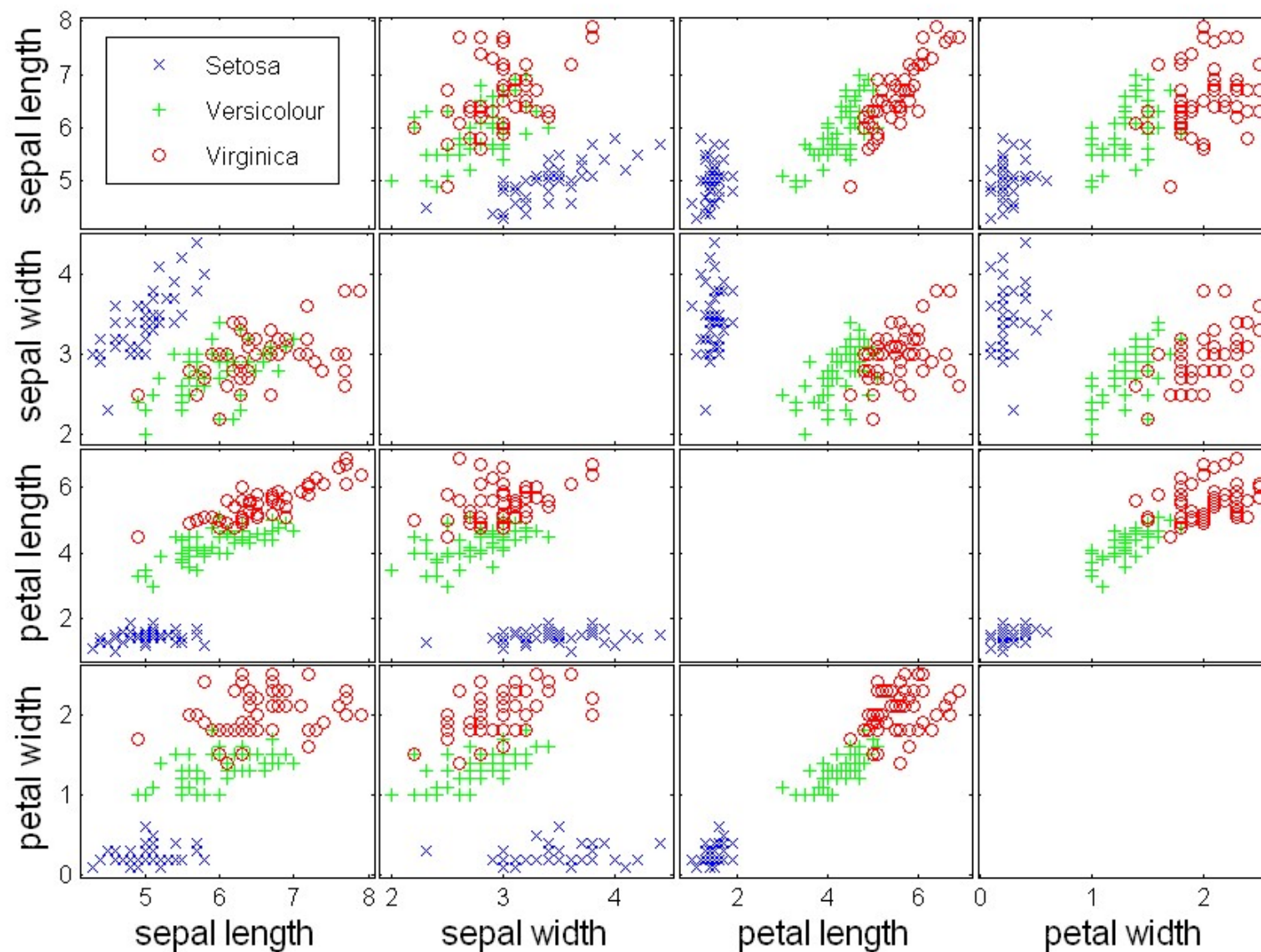
- **Box Plots**

- Another way of displaying the distribution of data



# Data Visualization (3)

## Scatter Plot Array



# RECOMMENDED BIBLIOGRAPHY

- Hands-On Machine Learning with Scikit-Learn and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems. Aurélien Géron. O'Reilly, 2019
- François Chollet. Deep Learning with Python, Manning, 2018. (on-line)
- Andrew Ng, Machine Learning Yearning, 2017.
- Tom Mitchell, Machine Learning. McGraw-Hill, 1997.
- <http://cs229.stanford.edu/>
- MOOC (Massive Open Online Courses)  
e.g. <https://www.coursera.org/>

# ANACONDA 3

**1) Install Anaconda 3 for Python 3:**

**<https://docs.anaconda.com/anaconda/install/>**

**2) Learn how to use Jupyter Notebook (part of Anaconda)**

**<https://www.dataquest.io/blog/jupyter-notebook-tutorial/>**

**Comment: If use higher versions than python 3.11 problems with tensorflow/ kerras libraries may arise.**

**Try to keep for now python version below 3.11.**